# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Developed **Java** server applications, integrating with **HSM**-protected cryptographic keys to support **device image signing** and factory provisioning workflows.
- Contributed to the architecture and development of services running on standalone on-premises servers with attached **HSMs**, ensuring high **security and assurance** requirements.
- Collaborated with device engineering teams to define and implement secure infrastructure system capabilities, advising on efficient and feasible cryptographic operations.
- Engineered **Python** scripts to automate data processing and reporting for **cryptographic key management** processes, reducing manual effort by **~30%**.
- Managed **production incidents** for enterprise applications using **Splunk** and **Dynatrace**, adhering to strict SLAs and performing root cause analysis for P1/P2 issues.
- Implemented **REST APIs** using **Java Spring Boot** for secure data exchange between internal services, including robust error handling and validation.
- Containerised Java applications using **Docker** for deployment on cloud infrastructure, ensuring consistency across development and production environments.

## SKILLS

**Languages** – Java, C, Python

**Frameworks/Libraries** – Spring Boot, RESTEasy, JAXB, JCA/JCE, OpenSSL, scikit-learn

**Security/Crypto** – HSM, PKI, Cryptography, Cryptographic Key Management, PKCS11, FIPS, Image Signing, Factory Provisioning

**Tools/Platforms** – Maven, Gradle, Git, Jenkins, Docker, Kubernetes, AWS (EC2, S3, Lambda, CloudWatch), Splunk, Dynatrace, PostgreSQL, MySQL, Oracle

**Concepts** – SDLC, CI/CD, REST API, XML, JSON, Embedded C, PowerPC, ELF Binary Generation, Architecture, Debugging, High Security Requirements

## PROJECTS

### Secure Key Management Service (HCL Technologies)

- Developed a **Java** service to manage cryptographic keys accessed via **HSM** integration, ensuring secure storage and retrieval for critical operations.
- Implemented **REST API** endpoints for key lifecycle management, including generation, rotation, and revocation, adhering to strict security protocols.
- Integrated with **PostgreSQL** for audit logging of all key operations, providing a detailed trail for compliance and security reviews.

### Automated Provisioning Pipeline (HCL Technologies)

- Created a **Python**-based pipeline to automate the provisioning of secure device configurations, integrating with factory systems.
- Designed and implemented shell scripts for managing deployments across multiple on-premises servers, enhancing operational efficiency.
- Utilised **Git** for source code management and **Jenkins** for continuous integration, ensuring reliable and repeatable build and deploy cycles.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)